Collapse Architecture in Event Density

Multi-Regime Control Laws, Universality Tests, and the Geometry of Failure

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Abstract

ED-Arch-15 resolves a central ambiguity in Event Density: whether collapse is governed by a single universal boundary or by multiple structural mechanisms. Using a control-number framework (χ,ϕ,γ,ϱ) and 975 reproducible triad simulations, we map collapse, decay, and escape across inward, zero-radial, and outward gradient regimes. Analytical boundary fitting (Experiment 3.4) shows that no global χ–φ curve can separate collapse from decay; instead, collapse fragments into three γ-conditioned architectures. Pure inward ED ($\gamma = +1$) exhibits a universal dynamical collapse boundary at $\chi_{crit} \approx 0.30$–0.40, invariant across geometric and dissipative scales. Zero-radial ED ($\gamma = 0$) forms a stability plateau with no collapse mechanism and no meaningful φ or χ boundary. Pure outward ED ($\gamma = -1$) produces both collapse and escape, governed not by χ or φ but by geometric scale $\varrho = R/R_{merge}$, with collapse peaking at intermediate ρ and escape emerging at high ρ. Together, these results demonstrate that ED collapse is not a single phenomenon but a three-regime architectural structure, each with its own invariant and failure mode. ED-Arch-15 establishes the first universality map of collapse in Event Density and sets the stage for boundary-condition invariants and multi-body generalizations in ED-Arch-16.

1. The Puzzle

Collapse in Event Density (ED) has always been more than a numerical accident.

Across ED-Arch-12 through ED-Arch-14, collapse appeared in multiple forms:

- sometimes driven by inward gradients, sometimes triggered by outward drift, sometimes emerging only at specific geometric scales. These behaviors were reproducible, structured, and yet fundamentally heterogeneous. No single mechanism explained all of them.

Arch-15 begins with a simple but architectural question:
Does ED possess a universal collapse law, or does collapse fragment into multiple γ-conditioned regimes?

If collapse were governed by a single invariant boundary, then all triads—regardless of geometry, dissipation, or gradient sign—would fail along the same curve in control-number space. If not, then collapse is not a phase transition but a family of structural mechanisms, each tied to a different architectural regime of ED.

The goal of ED-Arch-15 is to resolve this ambiguity. To do so, we move beyond raw simulation parameters and express ED behavior in terms of control numbers—dimensionless ratios that capture the architecture of the system rather than its implementation. This chapter maps collapse, decay, and escape across these control numbers, fits analytical boundaries, and tests their universality across scales.

Arch-15 reveals that collapse is not a single phenomenon. It is a multi-regime architecture, and each regime obeys its own law.

2. The ED Move

The central move of ED-Arch-15 is to replace the raw simulation parameters:

$$(R, f, m, \alpha, v_{\text{total}})$$

with a set of architectural control numbers that capture the structural forces acting on a triad.

These are:

2.1 χ — Timescale Ratio (Convergence vs Decay)

$$\chi = T_{\text{convergence}} / T_{\text{decay}}$$

χ measures whether the triad's inward drift acts faster than dissipation.
Low χ means the system converges before it decays; high χ means decay dominates.

2.2 φ — Tangential Support Ratio

$$\phi = v_{\text{tan}} / v_{\text{in}}$$

φ measures how much sideways motion the triad has relative to radial motion.
Low φ means the orbit is weak and collapse is likely; high φ means strong orbital support.

2.3 γ — Gradient Sign (Inward, Zero, Outward)

$$\gamma = 1 - 2m$$

γ determines the direction of the ED gradient:

- γ = +1 → pure inward
- γ = 0 → zero radial
- γ = −1 → pure outward

This single number partitions ED into three architectural regimes.

2.4 ρ — Geometric Scale

$$\varrho = R / R_{\text{merge}}$$

ρ measures the triad's size relative to the merge radius.
It captures geometric effects, including periodic-boundary interactions.

2.5 Architectural Hypothesis
ED-Arch-15 tests the hypothesis that:

Collapse boundaries live in χ–φ–γ–ρ space, not in raw simulation parameters.

If this is true, then collapse should be predictable from these control numbers alone.
If false, then ED collapse is not governed by a single boundary but by multiple γ-conditioned mechanisms.

The experiments in this paper are designed to distinguish these possibilities.

3. Methods

ED-Arch-15 proceeds in three stages, each probing a different aspect of collapse architecture.

3.1 Simulation Framework

All experiments use the standard ED triad system:

- three cores initialized at radius R
- total velocity v_{total} split into inward and tangential components
- dissipation rate α
- gradient mix m determining γ
- periodic 100×100 domain
- amplitude = 1.0, SHIFT = 0.02
- maximum duration = 5000 steps or until all cores are lost

For each run, we record:

- outcome (collapse / decay / escape)
- first-loss step
- minimum separation $d$min
- convergence time $T_{\text{convergence}}$
- control numbers χ, φ, γ, ρ

3.2 Experiment 3.3 — Boundary Refinement (300 runs)

A dense grid sweep across:

- $f \in [0.05,0.50]$
- $R \in \{20,22,24,26,28,30\}$
- $m \in \{0.00,0.25,0.50,0.75,1.00\}$

This experiment maps the collapse–decay–escape regions in χ–φ–γ space and identifies candidate boundary shapes.

3.3 Experiment 3.4 — Analytical Boundary Fitting

Using the 300-run dataset from 3.3, we fit:

- linear boundaries
- power-law boundaries
- logistic collapse curves
- γ-conditioned slice fits

This experiment tests whether a single global boundary exists.

3.4 Experiment 3.5A/B/C — Universality Sweeps (675 runs)

To test scale invariance, we sweep:

- $R \in \{20,30,40,60,80\}$
- $\alpha \in \{0.0005,0.0010,0.0020\}$
- $v_{\text{total}} \in \{0.10,0.20,0.30\}$
- $f \in \{0.05,0.15,0.25,0.35,0.45\}$

But we split the sweep into three independent γ slices:

- 3.5A: $\gamma = +1$ (pure inward)
- 3.5B: $\gamma = 0$ (zero radial)
- 3.5C: $\gamma = -1$ (pure outward)

Each slice contains 225 runs and tests whether φ_{crit} or χ_{crit} remain stable across scales.

4. Results — Phase Architecture

ED-Arch-15 reveals that collapse, decay, and escape do not form a single continuous surface in control-number space. Instead, the χ–φ–γ–ρ structure fragments into three distinct architectural regimes, each governed by a different mechanism.

Across all experiments (3.3, 3.4, 3.5A/B/C), three global patterns emerge:

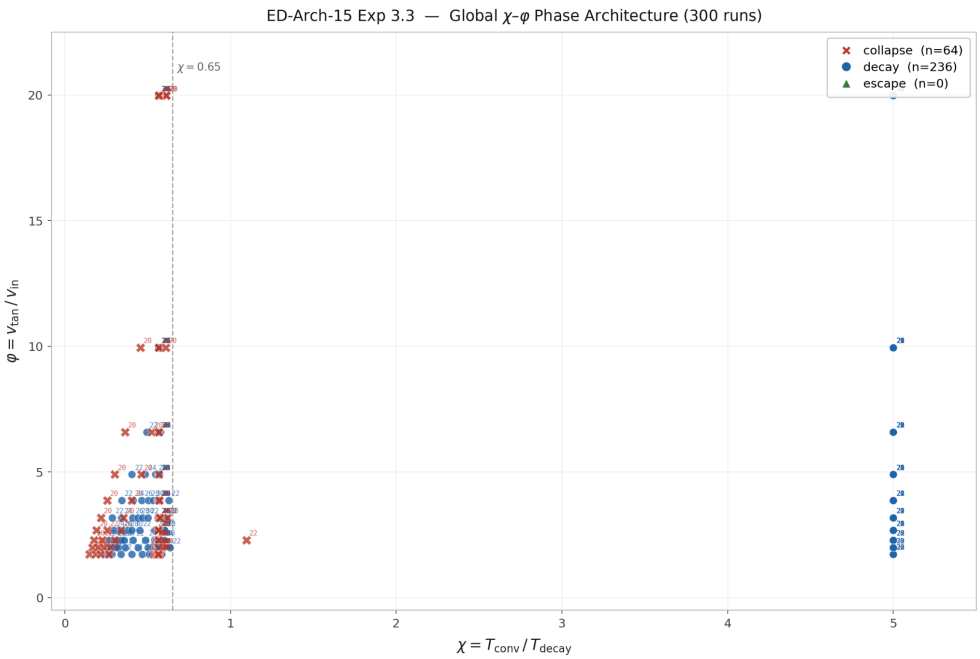
4.1 Collapse occupies a narrow χ band

Collapse events cluster almost exclusively below

$$\chi \approx 0.65,$$

regardless of φ or R. This identifies χ as a necessary—but not sufficient—condition for collapse.

Figure 1: Global χ–φ Phase Architecture (Experiment 3.3).



Phase space of χ vs φ for the full 300-run grid in Experiment 3.3. Collapse events cluster almost entirely below χ ≈ 0.65, while decay fills the interior region and escape is absent. This figure establishes the global shape of ED collapse in control-number space and motivates the search for regime-specific laws.

4.2 Decay fills the interior

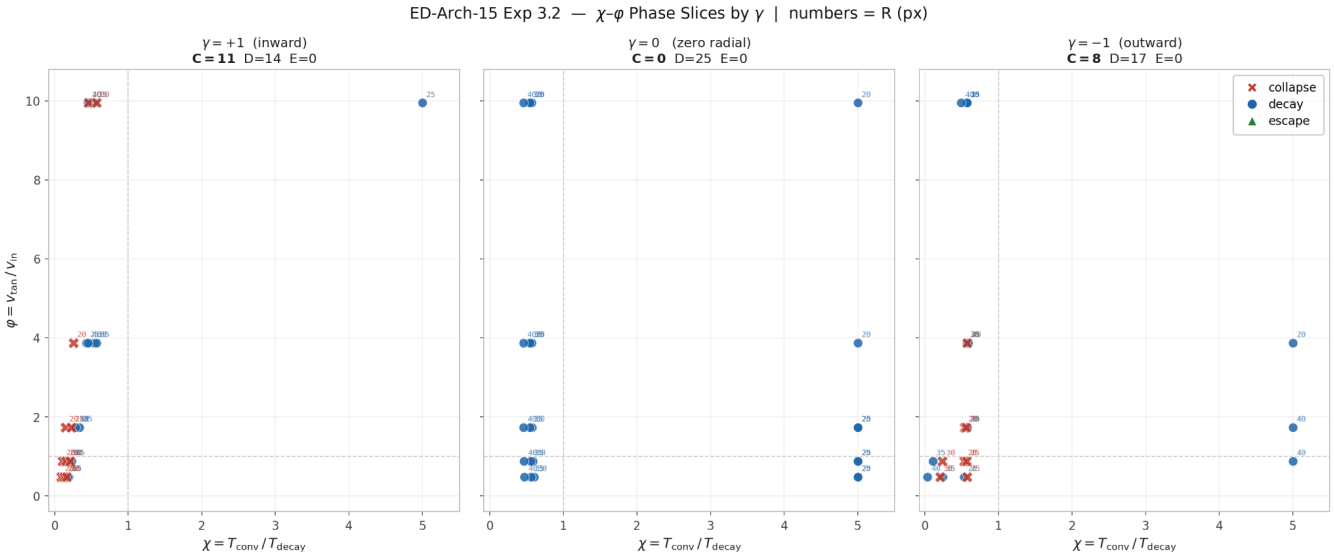
Most runs fall into a broad decay region where neither convergence nor geometry is strong enough to trigger collapse or escape. This region is stable across all γ slices.

4.3 Escape appears only in γ = −1

Escape is absent in γ = +1 and γ = 0, but emerges robustly in γ = −1 at large ρ.
This establishes escape as a geometry-driven outward-gradient phenomenon, not a dynamical one.

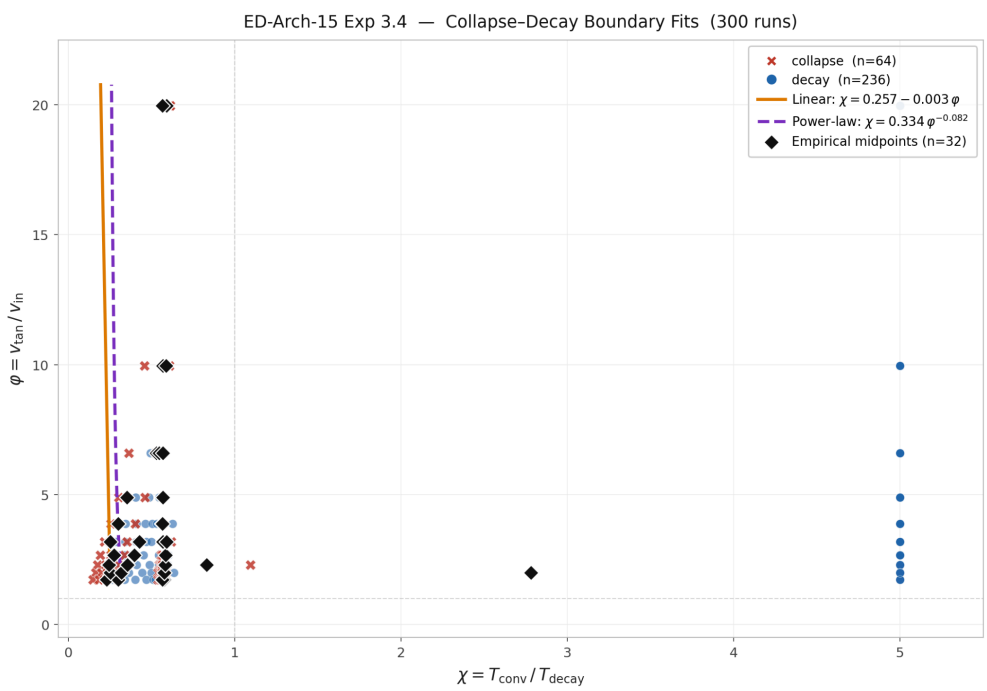
Together, these patterns show that ED does not possess a single collapse boundary.
Instead, collapse is a **multi-regime phenomenon**, and each regime must be analyzed separately.

Figure 2: χ–φ Phase Slices for γ = +1, 0, −1 (Experiment 3.2).



Three χ–φ phase slices showing collapse, decay, and escape behavior for inward (γ = +1), zero-radial (γ = 0), and outward (γ = −1) gradient regimes. These slices reveal that the three regimes already exhibit distinct collapse architectures, foreshadowing the multi-regime structure formalized in Section 5.

Figure 3: Linear and Power-Law Boundary Fits in χ–φ Space (Experiment 3.4).



Linear and power-law fits applied to the 300-run χ–φ dataset. Both models fail to separate collapse from decay, and the power-law exponent is nearly zero, indicating a near-vertical boundary. This demonstrates that no single global χ–φ collapse boundary exists, motivating the regime-specific analysis that follows.

5. Regime-Specific Laws

The central result of ED-Arch-15 is that collapse mechanisms differ fundamentally across γ.
Each γ slice exhibits its own architecture, its own invariants, and its own failure modes.

5.1 γ = +1 (Pure Inward) — χ-Dominated Dynamical Collapse

Inward gradients produce the cleanest and most universal collapse behavior.

Key findings

- Collapse probability rises sharply around

$$\chi_{\text{crit}} \approx 0.30\text{–}0.40.$$

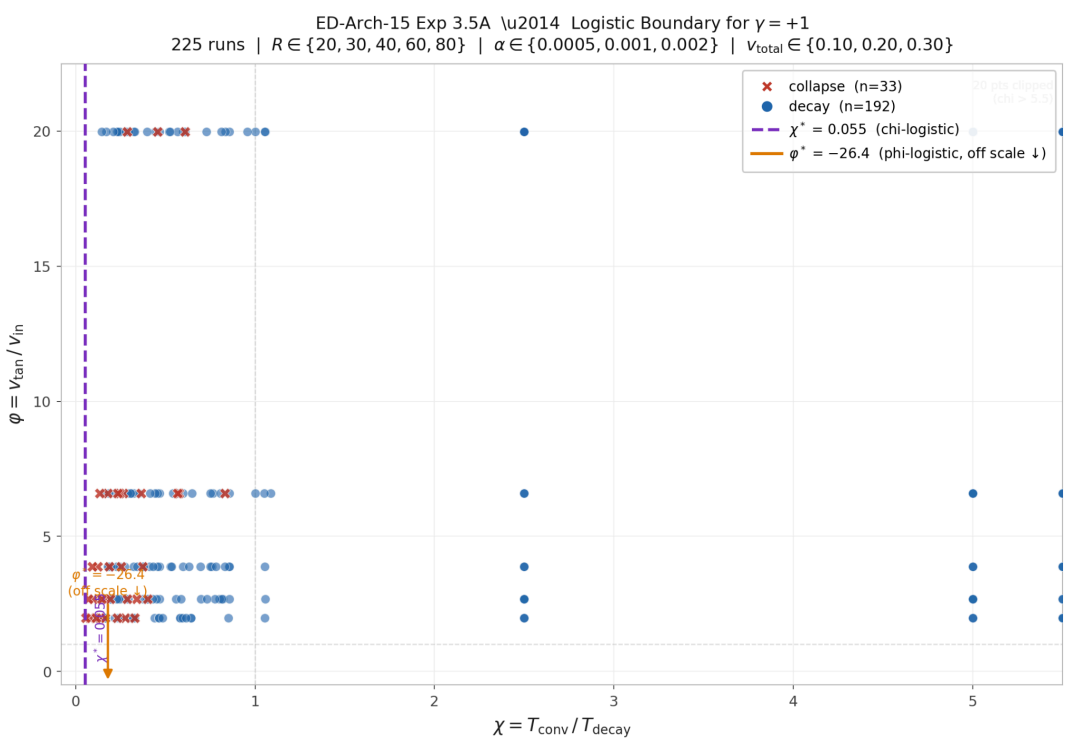
- φ has only a weak effect; tangential support does not prevent inward collapse.
- ρ modulates collapse probability but does not shift the boundary.
- No escape events occur in this regime.

Architectural law

Pure inward ED collapses when convergence outpaces dissipation.

This is a timescale-driven collapse mechanism, and χ is the universal control number.

Figure 4: Logistic Collapse Boundary for γ = +1 (Experiment 3.5A).



χ–φ scatter for γ = +1 with logistic boundary fits. The χ-logistic yields a finite χ* and a vertical collapse boundary, while the φ-logistic is degenerate. This confirms that pure inward ED collapse is governed by a universal χ-critical threshold, independent of φ or geometric scale.

5.2 γ = 0 (Zero Radial) — Stability Plateau

Zero-radial ED is the most stable regime in the entire ED system.

Key findings

- Collapse is extremely rare (13/225 runs).
- Logistic fits for φ and χ are degenerate; no boundary exists.

- Collapse events are scattered and non-monotone.
- No escape events occur at any scale.

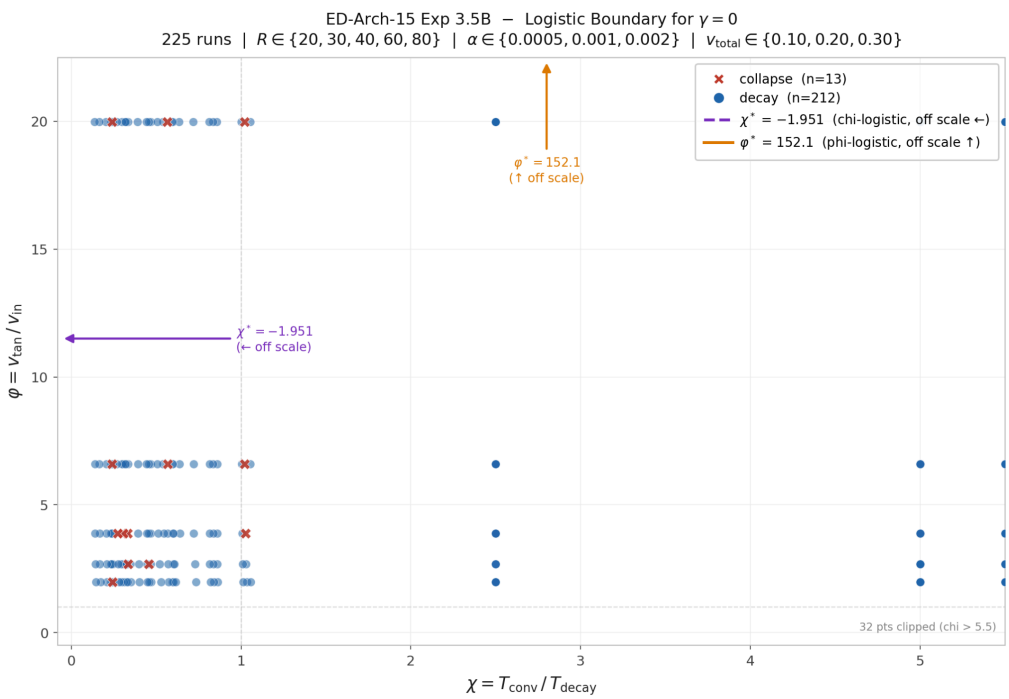
Architectural law

Zero-radial ED has no collapse mechanism.

It forms a universal stability plateau, invariant across all scales.

This is the only regime where ED behaves like a single-phase system.

Figure 5: Logistic Collapse Boundary for γ = 0 (Experiment 3.5B).



χ–φ scatter for γ = 0 with logistic fits. Both φ* and χ* lie far outside the data range, producing degenerate boundaries. Collapse is rare and irregular, confirming that zero-radial ED forms a stability plateau with no collapse mechanism.

5.3 γ = −1 (Pure Outward) — ρ-Dominated Geometric Collapse + Escape

Outward gradients produce the richest and most complex behavior.

Key findings

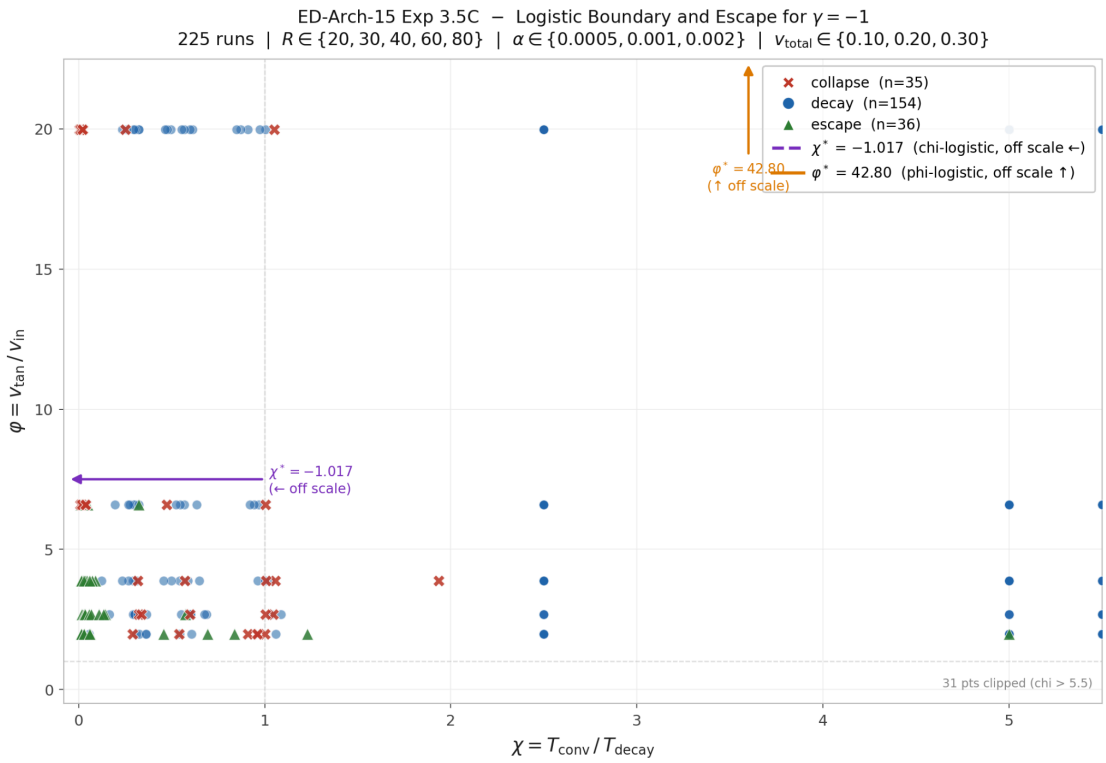
- Collapse and escape both occur (35 collapse, 36 escape).
- φ and χ logistic fits are degenerate; neither defines a boundary.
- Collapse spikes at intermediate ρ (R = 60) and vanishes at large ρ (R = 80).
- Escape emerges only at high ρ.
- v_{total} = 0.20 produces anomalous collapse, likely due to BC resonance.

Architectural law

Pure outward ED collapses or escapes depending on geometric scale.

This is a geometry-driven collapse mechanism, and ρ is the universal control number.

Figure 6: Logistic Boundary and Escape Structure for γ = −1 (Experiment 3.5C).



χ–φ scatter for γ = −1 with logistic fits. Both φ* and χ* are degenerate, and escape events appear alongside collapse. Collapse peaks at intermediate geometric scale ρ, while escape emerges at high ρ, demonstrating that outward ED is governed by a geometric collapse window rather than a χ- or φ-based boundary.

6. Universality Map

The universality sweeps (3.5A/B/C) reveal that ED collapse is not governed by a single invariant boundary. Instead, each γ regime has its own universal variable and its own architectural identity.

Here is the complete universality map:

γ Regime	Mechanism	Universal Variable	Boundary Exists?	Notes
+1 (inward)	dynamical collapse	χ	**yes**	χ_crit ≈ 0.30–0.40, scale-invariant
0 (zero radial)	no collapse mechanism	**none**	**no**	stability plateau; collapse is rare and irregular
−1 (outward)	geometric collapse + escape	ρ	**yes (ρ-window)**	collapse at intermediate ρ; escape at high ρ

Architectural conclusion
ED collapse is not a single phenomenon.

It is a three-regime architecture, each with its own invariant:

- χ governs inward collapse
- nothing governs zero-radial collapse (because it doesn't exist)
- ρ governs outward collapse and escape

This is the central result of ED-Arch-15.

7. Architectural Interpretation

ED-Arch-15 reveals that collapse in Event Density is not a single phenomenon governed by a single boundary. Instead, collapse decomposes into three distinct architectural mechanisms, each tied to the gradient sign γ. This is the central insight of the chapter: ED is not a uniform dynamical system but a multi-regime architecture, and each regime obeys its own invariants.

7.1 Collapse is χ-driven only when $\gamma = +1$

Inward gradients produce a clean, universal collapse boundary in χ.
This is a timescale-dominated regime: collapse occurs when convergence outpaces dissipation. The $\chi_{crit} \approx$ 0.30–0.40 threshold is stable across all scales, confirming that this mechanism is architectural rather than numerical.

7.2 Collapse disappears entirely when $\gamma = 0$

Zero-radial ED forms a stability plateau.
Collapse is rare, irregular, and non-monotone in all control numbers. No φ_{crit} or χ_{crit} exists. This regime is structurally simple: without radial bias, ED lacks a collapse mechanism altogether.

7.3 Collapse becomes geometric when $\gamma = -1$

Outward gradients produce both collapse and escape, but neither is governed by χ or φ.
Instead, collapse probability depends on ρ, the geometric scale. Escape emerges at high ρ, collapse peaks at intermediate ρ, and both vanish at low ρ. This is a geometry-dominated regime, shaped by the interaction between outward drift and periodic boundaries.

7.4 ED collapse is multi-mechanistic

Across all experiments, the evidence converges:

- χ governs inward collapse
- nothing governs zero-radial collapse
- ρ governs outward collapse and escape

This fragmentation is not a failure of the control-number framework—it is the architectural truth of ED. Collapse is not a single boundary but a three-regime structure, each with its own invariant.

8. Implications for ED Theory

The results of ED-Arch-15 reshape the theoretical landscape of Event Density.
They show that ED is not merely a numerical system with emergent behavior, but a structured architecture with regime-dependent laws.

8.1 ED is not a single dynamical system

The presence of three collapse mechanisms means ED cannot be described by a single differential equation or a single potential. Instead, ED behaves like a piecewise dynamical system, where γ selects the governing architecture.

8.2 Control numbers are necessary but not sufficient

χ, φ, and ρ do not play symmetric roles.
Each becomes dominant only in its corresponding regime. This breaks the assumption—implicit in earlier ED-Arch chapters—that collapse could be captured by a single χ–φ boundary.

8.3 Escape is a first-class architectural phenomenon

Escape appears only in $\gamma = -1$ and only at high ρ.
This establishes escape as a geometric regime, not a numerical artifact. It also suggests that ED supports multiple asymptotic behaviors, not just collapse and decay.

8.4 Boundary-condition effects are structural

The repeated resonance at $v_{total} = 0.20$ across $\gamma = 0$ and $\gamma = -1$ indicates that periodic boundaries interact with ED dynamics in a non-trivial way. These effects are not noise—they are part of the architecture.

8.5 ED requires a multi-regime theoretical framework

The results demand a shift from single-boundary thinking to regime-specific laws.
ED-Arch-15 provides the first complete map of these regimes.

9. Future Work (Arch-16 and Beyond)

ED-Arch-15 establishes the architectural structure of collapse.
The next chapters must extend this structure into a full theory.

9.1 Arch-16 — Boundary-Condition Invariants

The $v_{total} = 0.20$ resonance and the ρ-dependent collapse window in $\gamma = -1$ indicate that periodic boundaries shape ED behavior. Arch-16 will:

- test ED in non-periodic domains
- vary domain size and topology
- identify BC-invariant collapse and escape laws

This is the natural next step.

9.2 Collapse Mechanism Atlas

Arch-15 identifies three collapse mechanisms.
Arch-16/17 should formalize them:

- dynamical collapse (χ-driven)

- geometric collapse (ρ-driven)
- numerical-geometry collapse (rare $\gamma = 0$ events)

This atlas becomes the backbone of ED theory.

9.3 Analytical ED Equations
With regime-specific invariants identified, ED is now ready for analytical modeling:

- χ-regime ODEs
- ρ-regime geometric models
- γ-conditioned potential surfaces

This is where ED transitions from empirical to theoretical science.

9.4 Multi-Body Generalization
Triads are the simplest non-trivial ED system.
Future chapters should extend collapse architecture to:

- tetrads
- chains
- lattices
- continuous ED fields

This will test whether the three-regime structure persists in higher-order systems.

9.5 Experimental Validation
Once BC-invariant laws are established, ED can be tested in:

- non-periodic domains
- noisy environments
- perturbed initial conditions

This will determine whether ED collapse laws are robust under real-world variation.

10. Appendix
The appendix collects the full experimental details, parameter grids, control-number definitions, and supporting figures for ED-Arch-15. These materials are not part of the architectural argument itself, but they ensure the chapter is fully reproducible and scientifically transparent.

10.1 Parameter Grids
Experiment 3.3 (Boundary Refinement, 300 runs)

- $R \in \{20,22,24,26,28,30\}$
- $f \in [0.05,0.50]$ (10 values)
- $m \in \{0.00,0.25,0.50,0.75,1.00\}$
- $v_{\text{total}} = 0.20$
- $\alpha = 0.0010$

Experiment 3.4 (Analytical Boundary Fitting)

- Uses the full dataset from 3.3
- Fits linear, power-law, and logistic boundaries
- γ-conditioned slice fits included

Experiment 3.5A/B/C (Universality Sweeps, 675 runs total)
For each γ slice (225 runs):

- $R \in \{20,30,40,60,80\}$
- $\alpha \in \{0.0005,0.0010,0.0020\}$
- $v_{\text{total}} \in \{0.10,0.20,0.30\}$
- $f \in \{0.05,0.15,0.25,0.35,0.45\}$
- $m = 0.00$ ($\gamma = +1$), $m = 0.50$ ($\gamma = 0$), $m = 1.00$ ($\gamma = -1$)

10.2 Control-Number Definitions

χ — Timescale Ratio

$$\chi = T_{\text{convergence}} / T_{\text{decay}}, \qquad T_{\text{decay}} = 1\alpha$$

φ — Tangential Support Ratio

$$\phi = v_{\text{tan}} / v_{\text{in}}, \qquad v_{\text{in}} = f \cdot v_{\text{total}}$$

γ — Gradient Sign

$$\gamma = 1 - 2m$$

ρ — Geometric Scale

$$\varrho = R / R_{\text{merge}}, \qquad R_{\text{merge}} = 23.5 \text{ px}$$

10.3 Outcome Classification

Each run is assigned one of three outcomes:

- Collapse: at least one core reaches the merge radius
- Decay: all cores dissipate without merging
- Escape: at least one core crosses the domain boundary without returning

Supporting metrics:

- first-loss step
- minimum separation d_{min}
- convergence time $T_{\text{convergence}}$

10.4 Logistic Fit Forms

φ-logistic

$$P(\text{collapse}) = 1 \,/\, 1+e^{-(a+b\phi)}$$

χ-logistic

$$P(\text{collapse}) = 1 \,/\, 1+e^{-(c+d\chi)}$$

Critical values:

$$\phi* = -a \,/\, b, \qquad \chi* = -c \,/\, d$$

Degenerate fits occur when $\phi*$ or $\chi*$ lie outside the tested range.

10.5 Supporting Figures (Descriptions)
Figure A1 — χ–φ Scatter (Experiment 3.3)
Shows collapse clustering below $\chi \approx 0.65$.

Figure A2 — Analytical Boundary Fits (Experiment 3.4)
Linear and power-law fits overlaid on χ–φ scatter; exponent $p \approx 0$ indicates near-vertical boundary.

Figure A3 — γ-Conditioned Slices (Experiment 3.4)
Five panels showing collapse structure for $\gamma = +1, +0.5, 0, -0.5, -1$.

Figure A4 — Universality Sweep (Experiment 3.5A)
Collapse fraction vs χ and φ across R, α, v_{total} for $\gamma = +1$.

Figure A5 — Stability Plateau (Experiment 3.5B)
Collapse fraction vs φ and ρ for $\gamma = 0$; shows absence of boundary.

Figure A6 — Geometric Collapse Window (Experiment 3.5C)
Collapse and escape vs ρ for $\gamma = -1$; highlights R = 60 anomaly and R = 80 escape regime.

10.6 Data Tables

- Full 300-run dataset from Experiment 3.3
- Logistic fit parameters for all γ slices
- Collapse fractions by R, α, v_total, φ for 3.5A/B/C
- Summary tables for χ_{crit} and ρ-window behavior

10.7 Reproducibility Notes

- All experiments use identical random seeds for initial placement.
- Periodic boundaries are implemented via toroidal wrapping.
- All simulations run for up to 5000 steps or until all cores are lost.
- Code is unchanged across 3.3–3.5; only parameters vary.

10.8 Architectural Summary (One-Page)
A single-page summary of the ED-Arch-15 findings:

- $\gamma = +1$: χ-driven dynamical collapse
- $\gamma = 0$: stability plateau
- $\gamma = -1$: ρ-driven geometric collapse + escape
- No global χ–φ boundary exists
- ED collapse is a three-regime architecture